Surface Roller Model for `WAVE_OFFLINE` Coupling

Porting from WKB to Offline WEC

1 What is the surface roller?

When a wave breaks, not all of the breaking energy is instantly transferred to the mean current. A fraction α_r of the breaking dissipation ε^b feeds a *surface roller*—a mass of aerated water that rides on the front face of the broken wave and propagates shoreward at the wave phase speed c (not the group speed c_g). As the roller propagates, it dissipates at a rate ε^r that depends on the roller energy and wave properties. It is ε^r, not ε^b, that ultimately accelerates the mean nearshore current and drives breaking-induced turbulent mixing.

The physical consequence is a **shoreward displacement** of the effective momentum source: breaking occurs at the bar crest, but the roller carries that energy shoreward before releasing it. This broadens the cross-shore profile of the forcing and shifts the peak v location toward the shore (U2010, Fig. 6).

2 Roller energy balance

The roller action density is defined analogously to the primary wave action:

$$\mathscr{A}^r = \frac{E^r}{\sigma}, \tag{1}$$

where E^r is the roller energy density and σ is the intrinsic wave frequency. Its evolution equation (U2010 Eq. 40) is

$$\frac{\partial \mathscr{A}^r}{\partial t} + \nabla\cdot\left(\mathscr{A}^r\,\mathbf{c}\right) = \frac{\alpha_r\,\varepsilon^b - \varepsilon^r}{\sigma}, \tag{2}$$

where $\mathbf{c} = (\sigma/k^2)\,\mathbf{k}$ is the phase velocity vector (not the group velocity), and the two source/sink terms are:

- **Source:** $\alpha_r\,\varepsilon^b/\sigma$. A fraction $\alpha_r \in [0,1]$ of the primary wave breaking dissipation is converted into roller energy. The remainder $(1-\alpha_r)\varepsilon^b$ acts directly as breaking acceleration. Following U2010 Table 1, $\alpha_r = 1.0$ (all breaking feeds the roller) is the baseline; smaller values allow part of ε^b to bypass the roller.

- **Sink:** ε^r/σ. The roller dissipates at a rate (U2010 Eq. 41)

$$\varepsilon^r = \frac{g\sin\beta\;E^r}{c} = g\sin\beta\;\mathscr{A}^r\,k, \tag{3}$$

  where $\sin\beta \approx 0.1$ is an empirical roller slope parameter and $c = \sigma/k$ is the phase speed. The dissipation rate is proportional to the roller energy and to the local wavenumber—short, steep waves dissipate rollers faster than long, gentle ones.

Steady-state estimate

In a steady state ($\partial \mathscr{A}^r/\partial t = 0$) with weak advective divergence, the local balance gives

$$\varepsilon^r \approx \alpha_r \, \varepsilon^b, \tag{4}$$

i.e. the roller dissipation equals the source. The *total* effective dissipation entering the momentum equations is $\varepsilon^{\text{tot}} = (1-\alpha_r)\varepsilon^b + \varepsilon^r \approx \varepsilon^b$, confirming that the roller redistributes, but does not create or destroy, momentum. Its value lies in the spatial shift: the ε^r peak is displaced shoreward of the ε^b peak by the advection term $\nabla\cdot(\mathscr{A}^r\mathbf{c})$.

3 Modified breaking stress (U2010 Eq. 48)

Without the roller, the depth-averaged breaking acceleration is

$$\bar{\mathbf{B}}^b = \frac{\varepsilon^b}{\rho_0 \, \sigma \, D}\,\mathbf{k}. \tag{5}$$

With the roller, the momentum source becomes the *combined* dissipation (U2010 Eq. 50):

$$\boxed{\bar{\mathbf{B}}^b = \frac{(1-\alpha_r)\,\varepsilon^b + \varepsilon^r}{\rho_0 \, \sigma \, D}\,\mathbf{k}.} \tag{6}$$

In the code, this means replacing `ebrk` with `(1-alpha_r)*ebrk + rdsp` in the computation of `brk(i,j)`. With $\alpha_r = 1$, the direct breaking term vanishes entirely and all momentum comes through ε^r.

The 3D body force version (U2010 Eq. 48) includes a vertical shape function $f^b(z)$:

$$\mathbf{B}^b(z) = \frac{(1-\alpha_r)\,\varepsilon^b + \varepsilon^r}{\rho_0 \, \sigma}\,\mathbf{k}\, f^b(z), \qquad \int_{-h}^{\zeta} f^b(z')\,dz' = 1. \tag{7}$$

4 Modified breaking-induced mixing (U2010 Eq. 59)

The breaking-induced eddy viscosity K_v^b (added to KPP's K_v) is

$$\boxed{K_v^b = c_b \left(\frac{(1-\alpha_r)\,\varepsilon^b + \varepsilon^r}{\rho_0}\right)^{1/3} H_* \, D \, f^{K_v}(z),} \tag{8}$$

where $c_b \approx 0.03$ (U2010 Table 2, calibrated for DUCK94), $H_* = a_{K_v} H_s$ is a length scale based on the wave height, and $f^{K_v}(z)$ is a shape function (Type II in the code, decaying from surface). The exponent 1/3 comes from the TKE-based scaling $K \sim \ell\, q \sim H_* \, (\varepsilon/\rho_0)^{1/3}$.

In the code, this means replacing `ebrk` with `(1-alpha_r)*ebrk + rdsp` inside the Akb computation block. The shape function and c_b parameter are unchanged.

5 Roller Stokes drift

The roller also contributes to the Stokes drift. The roller Stokes transport (U2010 Eq. 42) is

$$\mathbf{U}^r = \frac{E^r}{\rho_0 \, \sigma}\,\mathbf{k} = \frac{\mathscr{A}^r}{\rho_0}\,\mathbf{k}, \tag{9}$$

so the total Stokes transport (U2010 Eq. 43) is

$$\mathbf{U}^{\mathrm{St}} = \frac{E + E^r}{\rho_0\, \sigma}\, \mathbf{k} = \frac{\mathscr{A} + \mathscr{A}^r}{\rho_0}\, \mathbf{k}. \tag{10}$$

The vertical distribution is assumed to follow the same profile as the primary Stokes drift (U2010 Eq. 44). In the code, this means multiplying the existing Stokes drift by $(1+\mathscr{A}^r/\mathscr{A})$ or, equivalently, replacing the wave action $\mathscr{A}$ with $\mathscr{A} + \mathscr{A}^r$ in the Stokes drift computation.

6 Advection scheme

The roller propagates at the wave **phase speed** $c = \sigma/k$, not the group speed. In the WKB code (`wkb_wwave.F`), the advection uses a 1st-order upwind scheme:

$$c^u_{i+1/2} = \tfrac{1}{2}\big(c^x_{i+1,j} + c^x_{i,j}\big), \tag{11}$$

$$c^v_{j+1/2} = \tfrac{1}{2}\big(c^\eta_{i,j+1} + c^\eta_{i,j}\big), \tag{12}$$

where $c^x = c\cos\theta$, $c^\eta = c\sin\theta$ are phase velocity components in the wave propagation direction. The upwind fluxes are

$$\mathcal{F}^x_{i+1/2} = \tfrac{1}{2}\,\Delta n_u\big[(c^u - |c^u|)\,\mathscr{A}^r_{i+1} + (c^u + |c^u|)\,\mathscr{A}^r_i\big], \tag{13}$$

$$\mathcal{F}^\eta_{j+1/2} = \tfrac{1}{2}\,\Delta m_v\big[(c^v - |c^v|)\,\mathscr{A}^r_{j+1} + (c^v + |c^v|)\,\mathscr{A}^r_j\big]. \tag{14}$$

The update is

$$\mathscr{A}^{r\,n+1}_{i,j} = \mathscr{A}^{r\,n}_{i,j} - \Delta t\, p_m\, p_n\big(\mathcal{F}^x_{i+1/2} - \mathcal{F}^x_{i-1/2} + \mathcal{F}^\eta_{j+1/2} - \mathcal{F}^\eta_{j-1/2}\big) + \Delta t\big(\alpha_r\, \varepsilon^b/\sigma - g\sin\beta\; \mathscr{A}^r\, k/\sigma\big). \tag{15}$$

CFL constraint

The upwind scheme is stable for $\mathrm{CFL} = c\,\Delta t/\Delta x \le 1$. In shallow water, $c \to \sqrt{gD}$. Representative values for $\Delta x = 30\,\mathrm{m}$, $\Delta t = 3\,\mathrm{s}$:

D (m)	c (m/s)	CFL	Stable?
2	4.4	0.44	yes
5	7.0	0.70	yes
10	9.9	0.99	marginal
20	14.0	1.40	**no**

For L4PTSAL, the surfzone (where rollers exist) typically has $D < 10\,\mathrm{m}$, so CFL < 1 is satisfied. In deeper water, $\varepsilon^b \to 0$, so the source term vanishes and $\mathscr{A}^r$ decays exponentially—the advection there is irrelevant. If needed, a CFL limiter or subcycling can be added for safety.

7 Connection to WW3 forcing

With `WAVE_OFFLINE`, the wave fields (H_s, σ, θ, ε^b, ε^{wd}, Q_b) are read from a WW3 forcing file and interpolated to each baroclinic time step. For the roller:

- ε^b (`wveb`): already read. This is the source term for the roller.

- Q_b (`wvqb`): fraction of breaking waves. Already declared in the code under `SURFACE_ROLLER` with the forcing reader (`nc_wwqb`). Available in the forcing file. Used for diagnostic initialization (see below), not needed for the prognostic equation itself.
- k, σ (`kw`, `fr`): computed from dispersion relation. Phase speed $c = \sigma/k =$ `fr/kw`.
- θ (`wdrx`, `wdre`): wave direction components. Phase velocity vector: $c^x = c\,$`wdrx`, $c^\eta = c\,$`wdre`.

Diagnostic initialization using Q_b

The roller can be initialized from a diagnostic steady-state estimate (Svendsen, 1984; U2010 Eq. 45):

$$E^r_{\text{diag}} = \rho_0\, g\, D\, A_R\, Q_b, \tag{16}$$

where $A_R = a^0_R\, H_s\, L_p$ is the roller area with empirical constants $a^0_R \approx 0.06$ and $L_p = 2\pi/|\mathbf{k}_p|$ is the peak wavelength. This avoids the cold-start transient from $\mathscr{A}^r = 0$.

8 Relationship to `SMOOTH_BRK`

`SMOOTH_BRK` (Laplacian smoothing of `brk`) is a *surrogate* for the roller: it spreads the breaking momentum isotropically over a Gaussian width $\sigma = \Delta x\sqrt{3N/8}$. The roller is the *physical* mechanism:

Property	`SMOOTH_BRK`	Roller
Direction	isotropic	shoreward (along $\mathbf{k}$)
Spreading width	tunable (N)	physical (β, c)
Conservation	exact (Neumann BC)	exact (PDE)
Akb modification	no	yes (Eq. 8)
Stokes drift	no	yes (Eq. 9)
CFL constraint	none	$c\,\Delta t/\Delta x < 1$
Complexity	$\sim$100 lines	$\sim$300 lines

With the roller implemented, `SMOOTH_BRK` becomes redundant and should be disabled (`#undef SMOOTH_BRK`). The two should not be used simultaneously, as that would double-smooth the forcing.